Insulation tester

The software of test devices enabled data transfer from the device to the computer. Today, field engineers can wirelessly stream measurement data to their phone, tablet or computer system using Bluetooth. Anritsu MT1000A, Hioki CM4374 and Testo 770-3 are among the equipment that sport Bluetooth facility.

Devices becoming cloud-connected. Field test devices can now be connected to a cloud server. Data collected from the field can be pushed to the server and retrieved by any connected device at any time.

Chhavda commented, "Previously, people needed PCs as an add-on with the T&M device to retrieve data from the device and generate reports. Now the devices have become cloud-connected, and in the process smartphone-connected. This has reduced the time of report generation, improved ease of data storage and increased the overall efficiency of operation."

For example, the Fluke Connect range of T&M devices feature auto-record and one-step transfer. So all the vital information is stored on the cloud, which can be accessed by any authorised team member. Quick analysis reports can be generated using TrendIt graphs.

Wherever data transfer and connectivity is involved, data protection becomes an important point of consideration. Madhukar Tripathi, senior manager, Anritsu India, said, "Field testing instrument must have various interfaces for data transfer. Nowadays data is critical to decide the next action. Therefore field testing instruments must have provision for protected data transfer."

Interactive interface. Touchscreens have found their way into handheld test equipment. Tripathi shared, "Due to the massive rise in the use of smartphones, all engineers or technicians today are used to touchscreen display. Nowadays, display or control of many handheld instruments can even be seen on a mobile device or tablet."

Manufacturers are improving the equipment software coupled with visual features like coloured display to further enhance the user interface.

Higher specs and combined functions. Field test equipment are improving in features, capacity and functionalities. Improved measurement capability and storage, optimised software and many-in-one functionality govern device upgrades.

For instance, "Earlier, some common instruments like clamp meters used to have a measurement capacity of up to 1000A. Now it has been upgraded to 2000A," shared a spokesperson from Hioki India. Talking about their own inventory of devices, he shared, "Our insulation tester has been upgraded with improved software compatibility for data logging and reporting. Power quality analysers have improved to accuracy level of 0.1 per cent."

Test equipment are being made compatible for up to 400G testing capacity. Anritsu's MT1100A harbours a dual-port 100G and 400G tester in a single unit.

Field engineers have to perform several tests, but carrying around separate instruments for different tests is a hassle. Keeping that in mind, manufacturers are designing equipment that are capable of multiple tests.